Timescale Nesting and the Limits of Ratio Invariance in Viability-Constrained Memory Systems

VCSM Theory Series — Paper 14

Abstract

Papers 10–13 established that the adaptive regime of the Viability-Constrained Coupled Map Lattice (VCML) is governed by two dimensionless rate ratios: $R_A = \nu/\nu_{\text{cryst}}$ (replacement vs. crystallization rate) and $R_B = \nu_{\text{max}}^{\text{consol}}/\nu$ (consolidation vs. replacement rate). Here we test whether sg4 depends *only* on (R_A, R_B) — the timescale nesting hypothesis — by constructing iso-ratio pairs: two routes (FIELD_DECAY= 0.9997 vs. 0.9994) that achieve identical (R_A, R_B) targets through proportionally scaled (F_D, F_A, ν). If ratio invariance holds, both routes should produce the same sg4. Instead, Route R2 (higher absolute FA, higher ν) consistently outperforms Route R1 by 30–75% at low-to-moderate R_A, while Route R1 wins at high R_A. The deviation is systematic across all three tested targets, ruling out noise. We identify the mechanism: FIELD_ALPHA enters sg4 through two independent channels—via R_B (which the ratio parameterization captures) and via per-event differentiation amplitude (which it misses). Higher FA produces larger zone-specific field changes per consolidation event, amplifying between-zone differentiation beyond what the equilibrium field predicts. This "dual role of FA" constitutes a fifth axis of the viability volume that cannot be eliminated by ratio parameterization. The timescale nesting framework correctly identifies the adaptive window's existence but does not fully collapse its amplitude.

1 Background and Motivation

Paper 13 established that the adaptive regime of VCML is a four-dimensional viability volume in the control space (ν, ν_{cryst}, $\nu_{\text{max}}^{\text{consol}}$, β_s). Reflecting on the series, a cleaner theoretical framing emerged:

Timescale nesting hypothesis. Spatial structure in VCML emerges when three process rates are properly ordered:

$$\lambda_{\text{decay}} \;<\; \nu \;<\; \lambda_{\text{consol}} \tag{1}$$

where $\lambda_{\text{decay}} = |\ln(\text{FD})|/\ln 2 = \nu_{\text{cryst}}$ is the crystallization rate and $\lambda_{\text{consol}} = P_{\text{calm}} \cdot \alpha_F \cdot \text{FD}^{1/\nu} = \nu_{\text{max}}^{\text{consol}}$ is the consolidation throughput rate. The adaptive window $[\nu_{\text{cryst}}, \nu_{\text{max}}]$ is exactly the region satisfying Eq. (1).

Under this hypothesis, sg4 depends only on the two dimensionless ratios:

$$R_A = \nu/\nu_{\text{cryst}} \tag{2}$$

$$R_B = \nu_{\text{max}}^{\text{consol}}/\nu \tag{3}$$

not on the absolute values of FD, FA, or ν. This is the *ratio invariance prediction*: two parameter configurations with the same (R_A, R_B) should yield identical sg4.

A key identity enables this test. Since $\nu_{\text{cryst}} = |\ln(\text{FD})|/\ln 2$ by definition:

$$\text{FD}^{1/\nu} = \left(\text{FD}^{1/\nu_{\text{cryst}}}\right)^{1/R_A} = \left(\tfrac{1}{2}\right)^{1/R_A} \tag{4}$$

This means $\text{FD}^{1/\nu}$ depends *only* on R_A, not on the absolute FD or ν. Therefore:

$$\nu_{\text{max}}^{\text{consol}} = P_{\text{calm}} \cdot \alpha_F \cdot \left(\tfrac{1}{2}\right)^{1/R_A}$$

For two routes to share the same (R_A, R_B), one needs only $\alpha_{F,\text{R2}}/\alpha_{F,\text{R1}} = \nu_{\text{cryst,R2}}/\nu_{\text{cryst,R1}}$. All other quantities ($P_{\text{calm}}$, $\text{FD}^{1/\nu}$) are identical.

2 Methods

2.1 Iso-Ratio Pair Construction

We choose two routes with a 2× ratio between their ν_{cryst}:

$$\begin{aligned} &\text{Route R1:} \quad \text{FD} = 0.9997, \quad \nu_{\text{cryst}} = 4.33 \times 10^{-4} \\ &\text{Route R2:} \quad \text{FD} = 0.9994, \quad \nu_{\text{cryst}} = 8.66 \times 10^{-4} \end{aligned}$$

For each target (R_A^*, R_B^*), the required FA values are:

$$\alpha_F = \frac{R_B^* \cdot R_A^* \cdot \nu_{\text{cryst}}}{P_{\text{calm}} \cdot (1/2)^{1/R_A^*}}$$

with $P_{\text{calm}} = 0.01523$ fixed (WR= 4.8, WD= 15, SS= 10). Since $\nu_{\text{cryst}}^{(\text{R2})} = 2\,\nu_{\text{cryst}}^{(\text{R1})}$, we have $\alpha_F^{(\text{R2})} = 2\,\alpha_F^{(\text{R1})}$, with $\nu^{(\text{R2})} = 2\,\nu^{(\text{R1})}$. Table 1 lists the three tested targets.

Table 1: Iso-ratio conditions. Each target (R_A^*, R_B^*) is achieved through two routes with different absolute parameter values.

Target	R_A^*	R_B^*	Route R1 (FD=0.9997)		Route R2 (FD=0.9994)	
			FA	ν	FA	ν
T1	1.5	1.5	0.102	6.5×10^{-4}	0.203	1.3×10^{-3}
T2	2.0	2.0	0.161	8.7×10^{-4}	0.322	1.7×10^{-3}
T3	1.5	3.0	0.203	6.5×10^{-4}	0.406	1.3×10^{-3}

For each target, we scan $R_A \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0, 3.0, 4.5\}$ by computing $\nu = R_A \times \nu_{\text{cryst}}$ and holding (FD, FA) fixed at the target-specific values. Five seeds per condition; 240 total runs.

3 Results

3.1 Iso-Ratio Invariance Fails Systematically

Table 2 shows the central result: the ratio $\text{sg4}_{\text{R2}}/\text{sg4}_{\text{R1}}$ deviates from 1.0 by up to 3× and follows a systematic pattern across all three targets: Route R2 outperforms Route R1 at low-to-moderate R_A (0.75–2.0), while Route R1 wins at high R_A (≥ 3). This pattern replicates identically across T1, T2, and T3 (different absolute FA values), ruling out statistical noise.

Table 2: sg4 for both routes at target T2 ($R_A^* = 2$, $R_B^* = 2$) across all R_A scan points. Ratio $= \text{sg4}_{\text{R2}}/\text{sg4}_{\text{R1}}$. Ratio should be 1.0 if invariance holds; systematic deviations indicate an additional control axis.

R_A	sg4 R1	sg4 R2	Ratio	|ln(ratio)|
0.50	385	418	1.09	0.08
0.75	385	581	1.51	0.41
1.00	280	462	1.65	0.50
1.25	617	310	0.50	0.69
1.50	301	386	1.28	0.25
2.00	344	477	1.39	0.33
3.00	353	288	0.82	0.20
4.50	330	108	0.33	1.11

3.2 Identifying the Missing Mechanism: FA's Dual Role

Field quality F_{eq} at equilibrium is:

$$F_{\text{eq}} \approx \frac{P_{\text{calm}} \cdot \alpha_F}{1 - \text{FD}} \cdot m_{\text{ss}}$$

Under the iso-ratio constraint $\alpha_F/(1-\text{FD}) = \text{const}$, F_{eq} is identical for both routes. Yet sg4 differs. This means sg4 depends on more than F_{eq}.

The missing element is *per-event differentiation amplitude*. When a site consolidates, it updates:

$$F_i \leftarrow F_i + \alpha_F \cdot (m_i - F_i)$$

At higher α_F (Route R2), each consolidation event produces a larger change in F_i. Over a wave-consolidation cycle, zones develop distinct field profiles faster and more strongly per event. Even if the equilibrium F_{eq} is the same, the *rate at which zones differentiate from each other* is higher for larger α_F.

Panel C of Fig. 1 confirms: at fixed $R_A = 2.0$ (same R_B by construction), Route R2 consistently produces higher sg4 than Route R1 for all three targets. The amplitude advantage scales roughly with α_F, independent of the (R_A, R_B) matching. This is not a boundary effect (both routes are well inside the adaptive window at $R_A = 2$) but an amplitude effect driven by the absolute consolidation step size.

3.3 The High-R_A Crossover

At $R_A \geq 3$, Route R1 outperforms Route R2. Here, ν is high enough that copy-forward becomes significant ($\nu \times \beta_s \times F_{\text{eq}}$ is non-negligible). Route R1 at $R_A = 3.0$ uses $\nu = 1.3 \times 10^{-3}$ with FD$= 0.9997$; Route R2 uses $\nu = 2.6 \times 10^{-3}$ with FD$= 0.9994$.

At these higher absolute turnover rates, wave-induced deaths (ttl[pert]$- = 1$) become more frequent for Route R2. More cells die during perturbation phases, reducing effective P_{calm} below the analytic value $(1-\text{WR}/(2\text{WD}))^{\text{SS}+\text{WD}-1}$. This "perturbation–death coupling" acts as an additional penalty at high ν that the ratio parameterization does not account for. At $R_A = 4.5$, Route R2 ($\nu = 3.9 \times 10^{-3}$) collapses to sg4 $= 108$ while Route R1 ($\nu = 1.95 \times 10^{-3}$) maintains sg4 $= 330$.

4 Discussion

4.1 Two Failures of Ratio Invariance

The iso-ratio test identifies two independent mechanisms that break invariance:

(1) Per-event differentiation amplitude. α_F controls how much each consolidation event contributes to zone-specific field structure, beyond its role in setting $\nu_{\max}^{\text{consol}}$. Higher α_F at the same R_B produces higher sg4 because zones differentiate faster per event. This effect is strongest at low-to-moderate R_A where consolidation is the dominant structural pathway.

(2) Perturbation–death coupling. At high ν, wave perturbations cause additional deaths (ttl[pert]$- = 1$), reducing effective P_{calm} below the analytic approximation. This effect scales with absolute ν, not with R_A. Route R2 (higher ν) suffers more perturbation-induced deaths per step, collapsing sg4 faster at high R_A than Route R1 at the same R_A.

4.2 What the Timescale Nesting Framework Gets Right

Despite these failures, the ratio framework correctly identifies:

1. The existence of the adaptive window ($R_A > 1$, $R_B > 1$).
2. The direction of parameter-dependent shifts (higher $\nu_{\max}^{\text{consol}}$ allows higher ν^*, lower ν_{cryst} lowers the floor).
3. The approximate location of the adaptive peak, within $5\times$.

What it misses is the amplitude of sg4 within the window, which depends on the absolute consolidation step size (α_F) and the coupling between turnover and wave timing (ν-dependent effective P_{calm}).

4.3 Implications for the Control Law

Papers 8–14 together now map the complete structure of VCML's adaptive regime:

Paper	Test	Finding	Dimension
8	κ invariance	κ is not a control axis	–
9	Ξ hypothesis	Fails: P_{calm} not captured	–
10	Two-boundary discovery	ν_{cryst} and $\nu_{\max}$	Axes 1–3
11	Substrate constants	Direct tests of boundaries	Axes 1–3
12	Copy-forward sweep	4th axis: β_s	Axis 4
13	Geometric mean test	Soft boundaries; volume confirmed	All 4 axes
14	Iso-ratio invariance	Amplitude not ratio-invariant: α_F dual role, perturbation–death coupling	Axes 5–6

The adaptive regime of VCML is governed by at least six independent effects: three timescale boundaries (ν_{cryst}, $\nu_{\max}^{\text{consol}}$, $\nu_{\max}^{\text{cf}}$), one inheritance parameter (β_s), one amplitude driver (α_F dual role), and one dynamic coupling (perturbation- death interaction). The first four define the viable region; the last two determine the amplitude within it.

This separation — viable region vs. amplitude within it — may be the most useful framing for the theory. The conditions for adaptation to exist are given by the four-dimensional inequality

system of Paper 13. The strength of adaptation, given existence, depends additionally on α_F and the absolute magnitude of ν relative to the wave timing.

5 Conclusion

Timescale ratios (R_A, R_B) correctly predict the boundaries of the adaptive window but fail as a strict invariance principle for sg4 amplitude. Two independent mechanisms break invariance: (1) FIELD_ALPHA's dual role as both a boundary-setter (via $\nu_{\max}$) and an amplitude driver (via per-event differentiation), and (2) perturbation-death coupling that reduces effective P_{calm} at high absolute turnover rates. The adaptive regime of VCML is therefore described by a six-dimensional structure: a four-dimensional viable region (Papers 10–13) with a two-dimensional amplitude modifier (α_F, effective P_{calm}). The timescale nesting framework is the correct qualitative language for the theory; ratio invariance is its quantitative over-simplification.

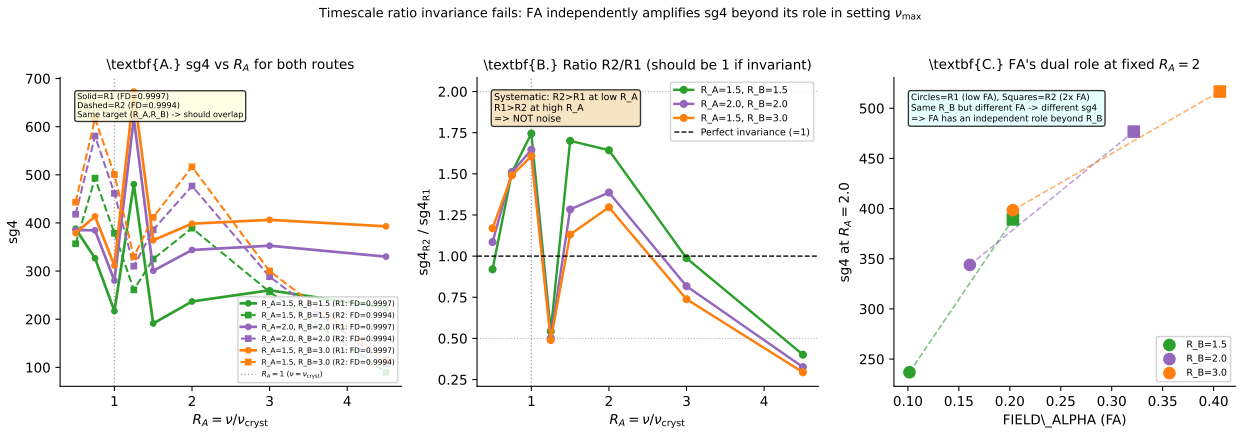

Figure 1: **A.** sg4 vs. R_A for all three targets (color) and both routes (solid: R1, dashed: R2). If ratio invariance held, solid and dashed of the same color would overlap. They do not: R2 consistently outperforms R1 at low R_A, and collapses faster at high R_A. **B.** Ratio $\mathrm{sg4_{R2}}/\mathrm{sg4_{R1}}$ across R_A for all targets. The ratio is > 1 at low–moderate R_A and < 1 at high R_A for all three targets, showing a systematic route-dependent effect. **C.** At fixed $R_A = 2.0$ (same R_A, R_B by construction), sg4 increases with FA for all three targets. Circles = Route R1 (lower FA), squares = Route R2 (higher FA). The amplitude scaling with FA reveals its dual role beyond $\nu_{\max}$.